

Study Report: Introduction to Machine Learning with Python - ML / AI Engineer

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Family: ML / AI Engineer

Level: Beginner

Chapters: 7

Source: Reference: Introduction to Machine Learning with Python - Andreas Müller & Sarah Guido

Official sources and free libraries

- <https://www.oreilly.com/library/view/introduction-to-machine/9781449369880/>

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Book_Reports/06-machine-learning/study-report-introduction-to-machine-learning-with-python-a-guide-for/report.json

Summary

Study report for *Introduction to Machine Learning with Python* by Andreas Müller & Sarah Guido (Beginner level) mapped to the ML / AI Engineer role. scikit-learn workflow: preprocessing, models, and evaluation.

Chapters

Track: Book-Intro

Chapter 1: About Introduction to Machine Learning with Python [Beginner]

Chapter focus: About Introduction to Machine Learning with Python** *(Level: Beginner)

This study report summarizes *Introduction to Machine Learning with Python* by Andreas Müller & Sarah Guido for the ML / AI Engineer role. The resource is rated Beginner level. scikit-learn workflow: preprocessing, models, and evaluation. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Introduction to Machine Learning with Python
# Author: Andreas Müller & Sarah Guido
```

```
# Role: ML / AI Engineer
# Level: Beginner
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Introduction to Machine Learning with Python.

When to use it

When learning ML / AI Engineer skills at Beginner level.

Where to use it

ML / AI Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Beginner]

Chapter focus: Why This Book Matters for Your Role** *(Level: Beginner)

ML / AI Engineer professionals use ideas from Introduction to Machine Learning with Python to solve real workplace problems. scikit-learn workflow: preprocessing, models, and evaluation. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: ML / AI Engineer
Book focus: scikit-learn workflow: preprocessing, models, and evaluation.
Recommended level: Beginner
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

ML / AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a beginner skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Beginner]

Chapter focus: Key Topics Covered** *(Level: Beginner)

The main topics in Introduction to Machine Learning with Python include practical concepts described as: scikit-learn workflow: preprocessing, models, and evaluation. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in ML / AI Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: ML/AI Engineer Role in 2024-2026 [Beginner]

Chapter focus: ML/AI Engineer Role in 2024-2026** *(Level: Beginner)

ML engineers ship models to production - not just notebook accuracy. You own features, training, evaluation, deployment, and monitoring. LLM tooling augments feature work but classical ML remains core for tabular prediction, ranking, and fraud.

Code Reference:

```
from sklearn.pipeline import Pipeline
```

```
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

pipe = Pipeline([('scale', StandardScaler()), ('clf', LogisticRegression())])
```

What it shows: Pipeline chains preprocessing and model - prevents train-serve skew.

Topic guide

What it actually is

An ML engineer builds and operates predictive systems in production.

When to use it

When rules-based systems fail at scale or personalization is needed.

Where to use it

Recommendations, churn, fraud, demand forecast, and ed-tech risk scoring.

Real use example

CodeQuest predicts which students need nudges - you own the weekly retrain job.

Chapter 5: Applied: scikit-learn Pipelines [Beginner]

Chapter focus: scikit-learn Pipelines** *(Level: Beginner)

Pipelines bundle preprocessing and estimators. Fit on train only; transform val/test to prevent leakage. ColumnTransformer applies different steps per column type.

Code Reference:

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer

num_pipe = Pipeline([('impute', SimpleImputer()), ('scale', StandardScaler())])
preprocess = ColumnTransformer([('num', num_pipe, num_cols), ('cat', OneHotEncoder(),
cat_cols)])
```

What it shows: ColumnTransformer keeps numeric and categorical handling explicit.

Topic guide

What it actually is

sklearn Pipelines standardize reproducible ML workflows.

When to use it

Every tabular ML project before custom deep learning.

Where to use it

Marketing uplift, credit scoring, and maintenance prediction.

Real use example

One Pipeline.pkl deployed to API - training and serving use identical transforms.

Chapter 6: Applied: Train/Validation/Test Splits [Beginner]

Chapter focus: Train/Validation/Test Splits *(Level: Beginner)**

Holdout test set is touched once. Validation tunes hyperparameters. Use stratified splits for imbalanced classes; TimeSeriesSplit for temporal data.

Code Reference:

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y,
random_state=42)
```

What it shows: stratify preserves class ratio in both splits.

Topic guide

What it actually is

Proper splits estimate real-world generalization.

When to use it

Before any metric comparison or model selection.

Where to use it

All supervised learning projects.

Real use example

Tuning on test by mistake inflated AUC - validation split catches overfitting.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Beginner]

Chapter focus: Study Plan and Practice *(Level: Beginner)**

Finish Introduction to Machine Learning with Python with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to ML / AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a beginner project screenshot.